Signal Detection Experiment
PSY310 - Lab in Psychology
Lab Report

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GitHub Link: https://github.com/siddharthap2400028/signal-detection

Introduction

In Psychophysics, Signal Detection Theory (SDT) is a psychological method employed to study perceptual decision-making by humans and other animals. It involves distinguishing one signal from other stimuli or background noise, particularly when a stimulus is not easily detected or identified and must be judged as present or absent (Green & Swets, 1966; Macmillan & Creelman, 2005). Performance may result in a hit, miss, false alarm, or correct rejection. A hit occurs when a present signal is detected correctly, while a miss occurs when the signal is present but is not detected. A false alarm occurs when a participant identifies a signal as present when it is actually absent, whereas a correct rejection occurs when the participant correctly identifies that the signal is absent (Stanislaw & Todorov, 1999). SDT also assesses performance through sensitivity (d′), which represents the ability to distinguish the signal from noise, and response criterion (c), which reflects the tendency to respond in a particular way (Macmillan & Creelman, 2005; Stanislaw & Todorov, 1999).

Method

Participant

The participant was an undergraduate student (18 years old) of Ahmedabad University. This experiment was conducted with informed consent and the participant was told that they would be shown gratings to identify whether they were tilted or straight.

Materials and Procedure

This experiment was designed and executed with PsychoPy 2025.1.1 on a desktop computer (1920 x 1200 pixel screen). Tasks included 100 trials, which tested the participant's ability to decide the tilt or lack of tilt of a visual grating.

The experimental trials started with a cross fixation displayed for 500 milliseconds. A Gaussian mask was then applied to a sinusoidal grating and the latter was left on the screen for 300 milliseconds. The grating was presented at random with the grating tilted to the left, to the right or straight. Trials were performed with a tilt of +5°, +1°, −1° or −5° in the tilted trials, with 0° in the straight trials. The participant was asked to click the up-arrow key if the grating was straight, and the down-arrow key if it was tilted. All responses were entered into a psychoPy program and then copied to an excel spreadsheet for analysis.

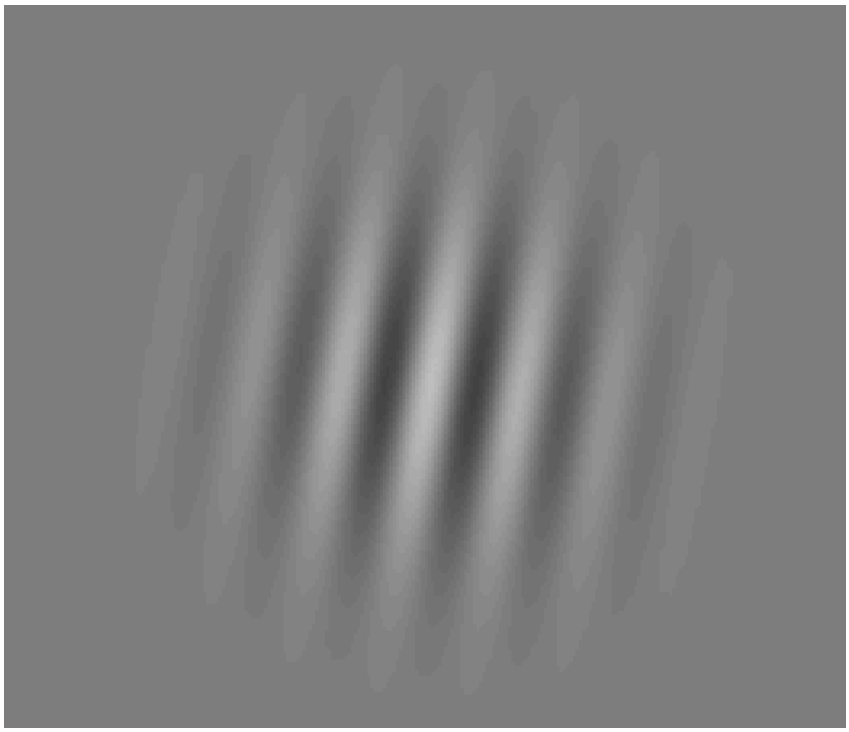


Figure 1 - Sinusoidal grating with a Gaussian mask, tilted 5 degrees rightward.

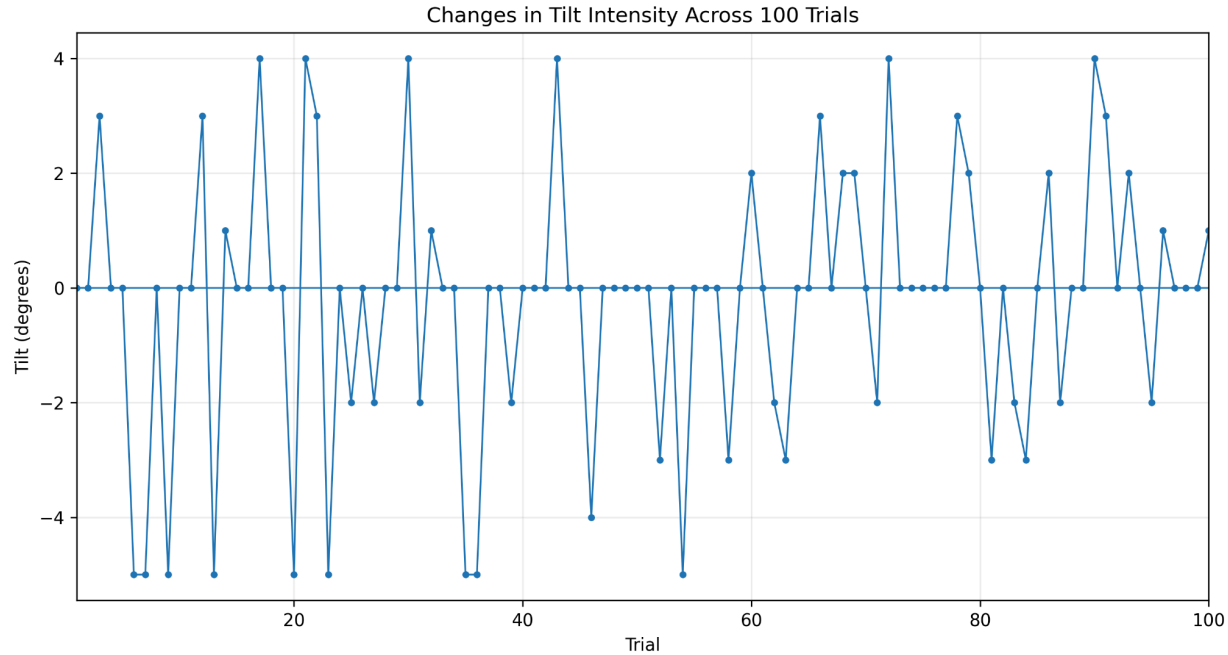


Figure 2 - The changes in tilt intensity from the first to the hundredth trial.

Results

The subject achieved 40 correct rejections, 14 false alarms, 38 hits and 8 misses during 100 trials. The hit rate was 82.6%, with a false alarm rate of 25.9%. The accuracy overall was 78%. The derived sensitivity was d′ = 1.58 and the response criterion was c = −0.15. The d' value shows fairly good sensitivity to distinguish between tilted gratings and straight gratings and the slightly negative criterion shows a small tendency to respond that a tilt was present.

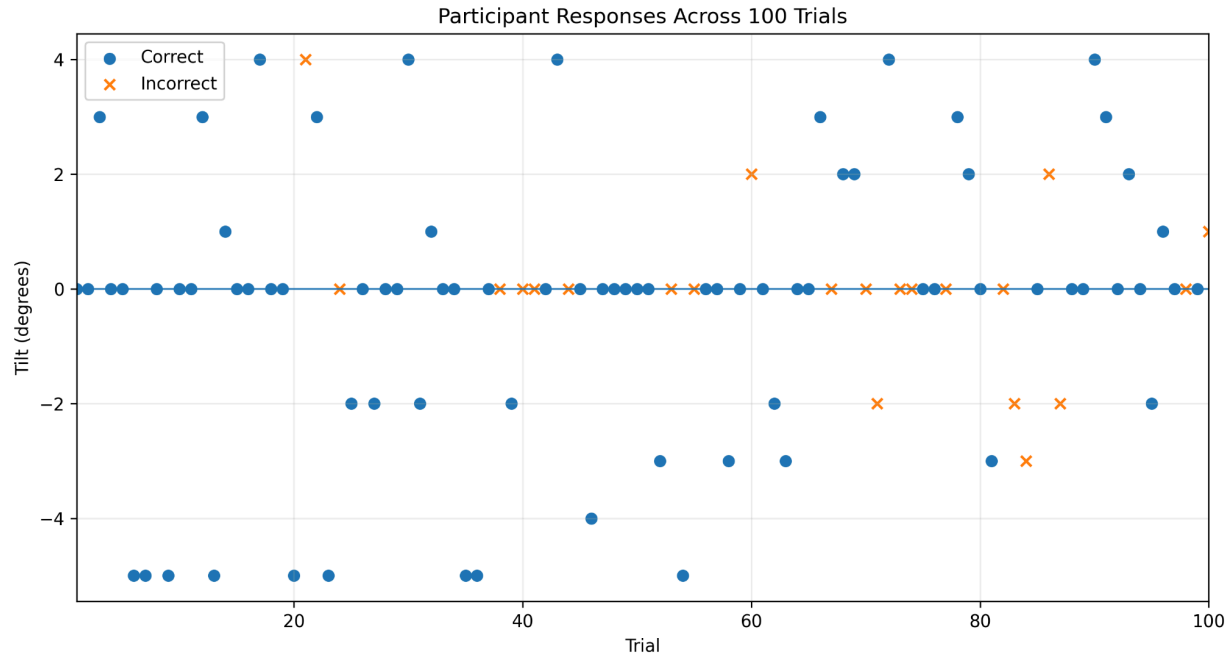


Figure 3 - Participant's accuracy over the 100 trials.

Discussion

The participant was 78% accurate with 100 trials overall. The value of d′=1.58 suggests a fairly good sensitivity to distinguish between tilted and straight gratings. The −0.15 criterion indicates a small amount of responding that a tilt was present. Higher tilts were more easily perceived and smaller tilts less easily perceived. Overall, the findings indicate that the participant was able to recognize the tilted and straight gratings with an acceptable level of accuracy, albeit this accuracy was influenced by the difference in the tilted and straight gratings.

References

Green, D. M., & Swets, J. A. (1966). *Signal detection theory and psychophysics*. Wiley.

Macmillan, N. A., & Creelman, C. D. (2005). *Detection theory: A user's guide* (2nd ed.). Lawrence Erlbaum Associates.

Stanislaw, H., & Todorov, N. (1999). Calculation of signal detection theory measures. *Behavior Research Methods, Instruments, & Computers, 31*(1), 137–149. https://do9i.org/10.3758/BF03207704